METRIC
MIL-DTL-24705/4A
27 May 2014
SUPERSEDING
MIL-DTL-24705/4
23 August 2010

DETAIL SPECIFICATION SHEET

PENETRATORS, MULTIPLE CABLE, COMPACT RECTANGULAR METAL FRAME AND BLOCKS, ELECTRICAL AND FIBER OPTIC CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-24705.

COMPACT RECTANGULAR CABLE PENETRATOR.

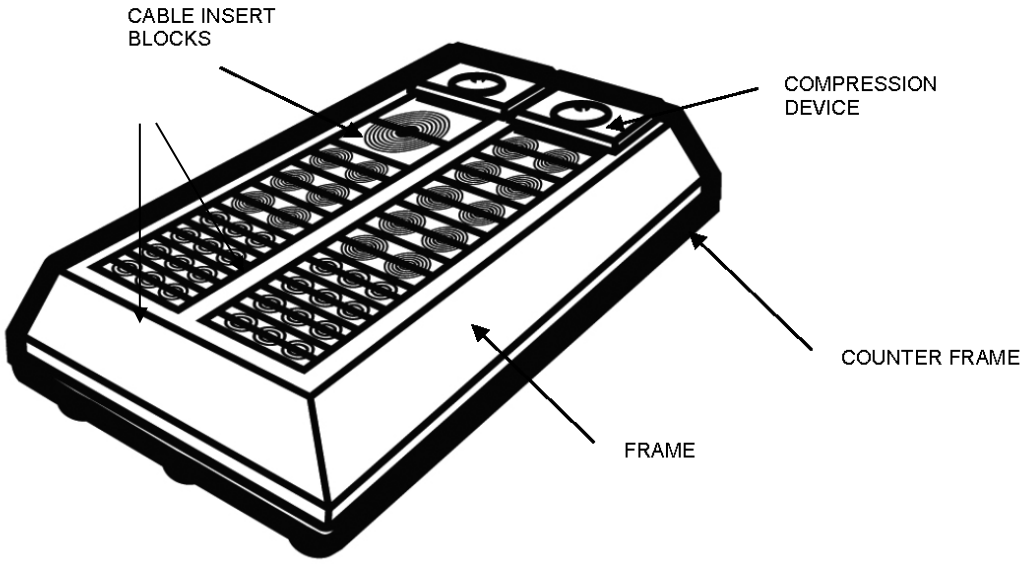
Intended use. Compact rectangular metal frame, cable insert blocks, and filler insert blocks are intended for use with fiber optic cables as specified in MIL-PRF-85045 and interconnection boxes as specified in MIL-DTL-24728/9, MIL-DTL-24728/10, electrical enclosures, and electronic equipment enclosures having cutouts of 215.9±2.5 mm (8.5±0.1 inches) by 110±2.5 mm (4.33±0.1 inches). The compact rectangular metal frame, cable insert blocks, and filler insert blocks specified by this specification sheet shall not be used to seal bulkheads or deck penetrations. For space within a frame where no cable entry is needed, BN20s shall be used to the maximum extent possible.

Compact rectangular cable penetrator assembly (figure 1). Components required for each compact rectangular cable penetrator specified by this specification sheet shall be as follows:

a. One compact rectangular metal frame assembly furnished with associated hardware (see figure 2).
   (1) Metal frame
   (2) Compression devices (quantity: 2)
   (3) Counter frame
   (4) Hexagon nuts (quantity: 10)
   (5) O-ring
   (6) Threaded inserts (quantity: 10)

b. Compact cable insert blocks (see figures 3, 4, and 5) and filler insert blocks (see figure 6) to fill the interior of the compact rectangular metal frame assembly. Blocks are furnished separately.

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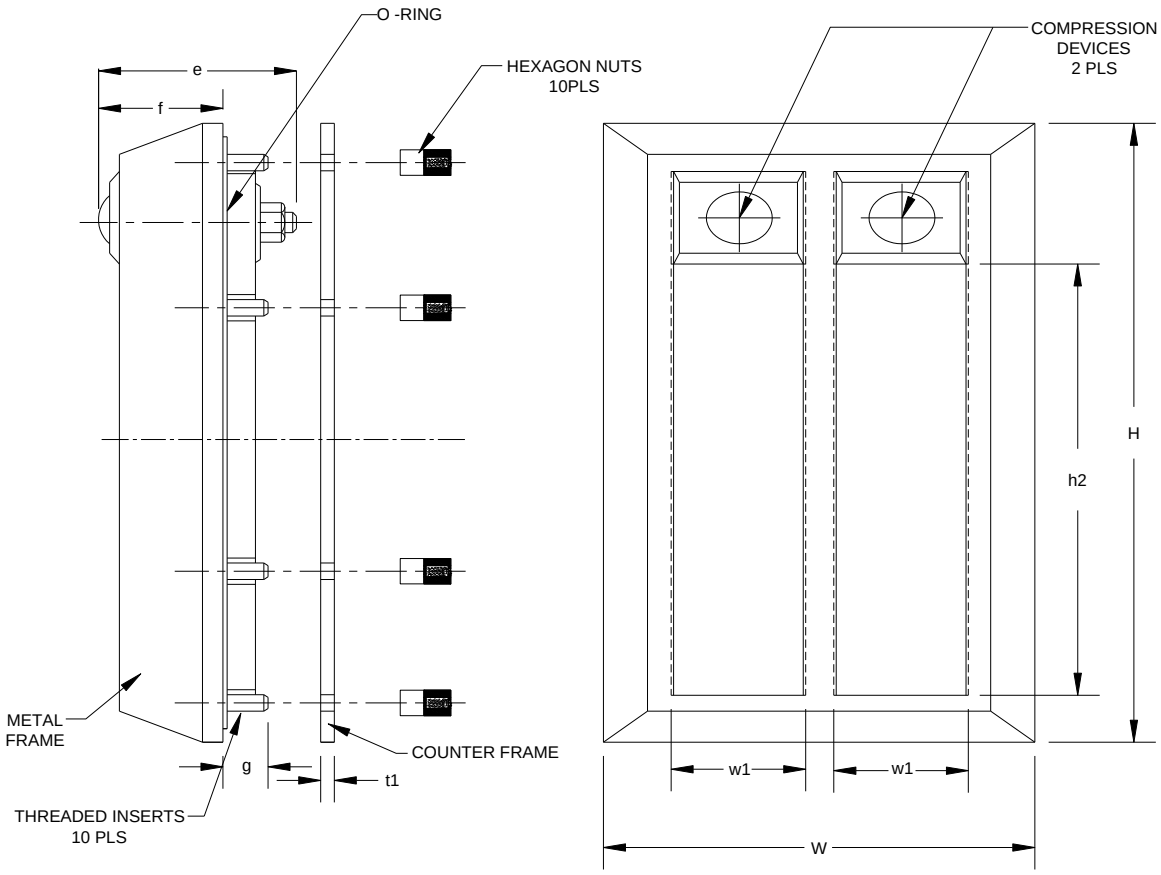
NOTES:

1. Drawing is not to scale.
2. Tolerances for molded rubber parts shall be in accordance with the Rubber Manufacturers Association (RMA) Drawing Designation A3 Commercial tolerance using guidance specified in MIL-HDBK-149, in the table entitled “Standard dimensional tolerances - molded solid rubber products rubber manufacturers association (RMA) (69)”.

FIGURE 1. Compact rectangular cable penetrator – assembled view – typical.

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Compact rectangular frame and associated hardware. Compact rectangular frame and hardware shall be in accordance with figure 2.



Item type	Item material	Item designation	e	f	g	h2	t1	w1	W	H
F	A	01	59.0 (2.32)	38.0 (1.50)	1/	160.0 (6.30)	4.00 (0.16)	40.0 (1.58)	130.0 (5.12)	230.0 (9.06)

NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Single decimal place tolerances are ±2.5 mm (±0.098 inch).
5. Two decimal place tolerances are ±0.38 mm (±0.015 inch).

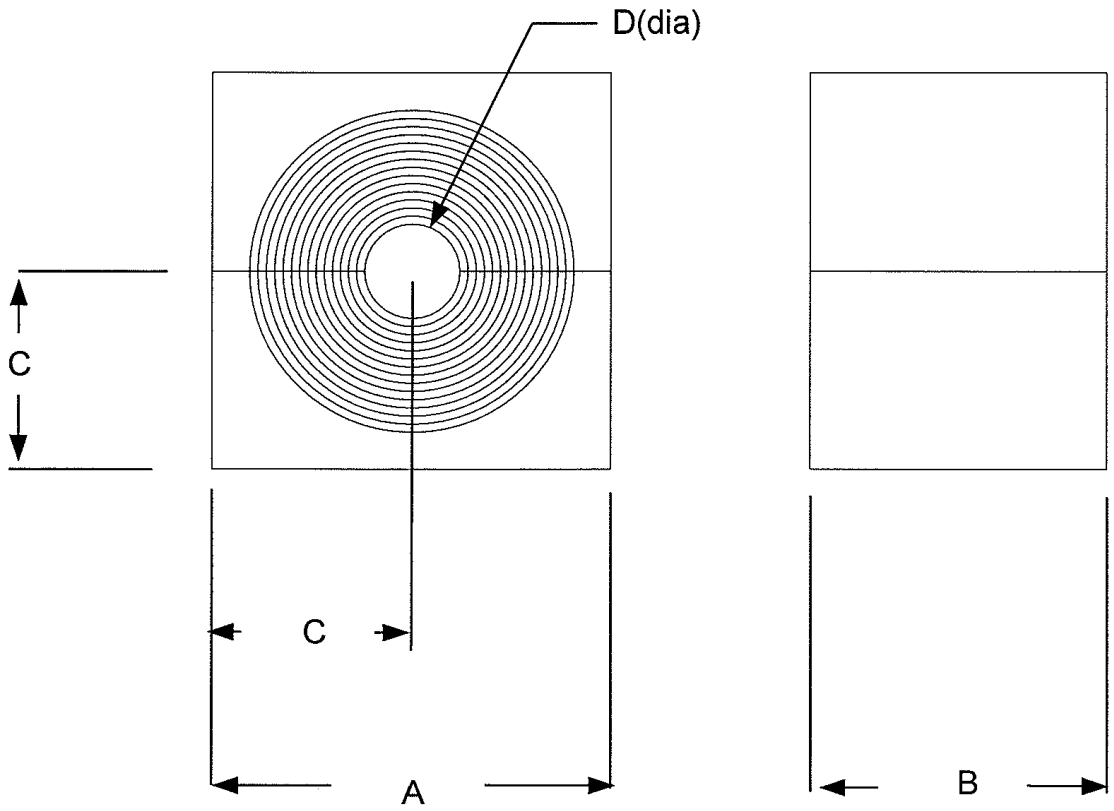
FOOTNOTES:

1/ Frame can be mounted in enclosures that have a panel thickness range of 3.0 – 7.0 mm (0.12 – 0.28 inch).

FIGURE 2. Compact rectangular metal frame and associated hardware.

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Compact two-piece square-faced one-cable insert block. Compact two-piece square-faced one-cable insert blocks shall be in accordance with figure 3.



Item type	Item material	Item designation	A	B	C	D
B	N	40	40.0 (1.57)	30.0 (1.18)	20.0 (0.79)	10.0 (0.39)

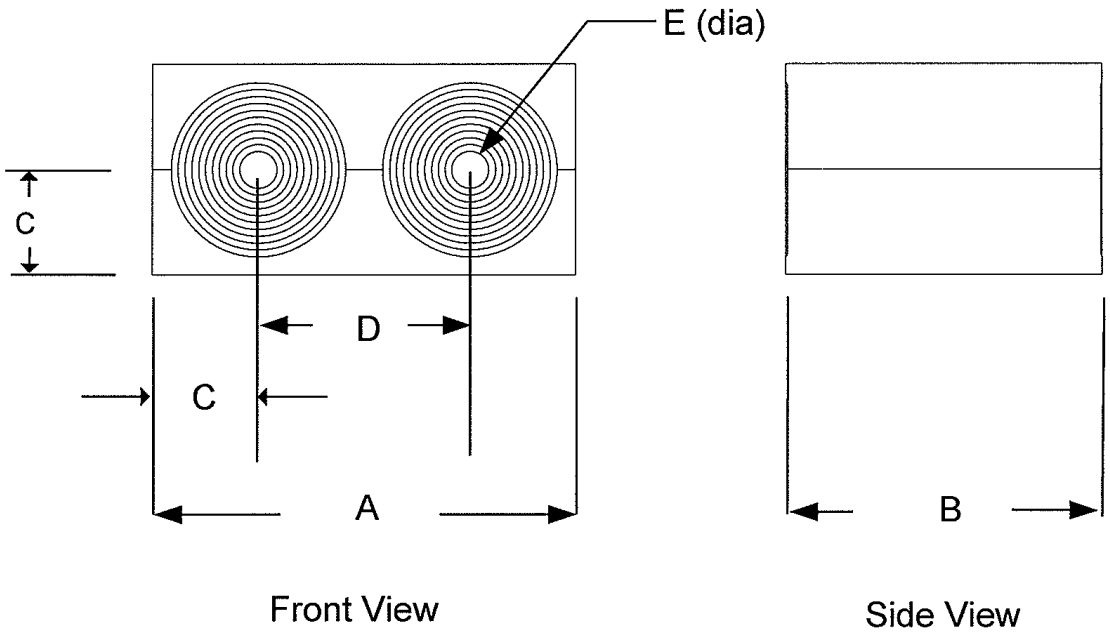
NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 3. Compact two-piece squared-faced one-cable insert block.

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Compact two-piece rectangular-faced two-cable insert block. Compact two-piece rectangular-faced two-cable insert blocks shall be in accordance with figure 4.



Item type	Item material	Item designation	A	B	C	D	E
B	N	20	40.0 (1.57)	30.0 (1.18)	10.0 (0.39)	20.0 (0.79)	3.2 (0.13)

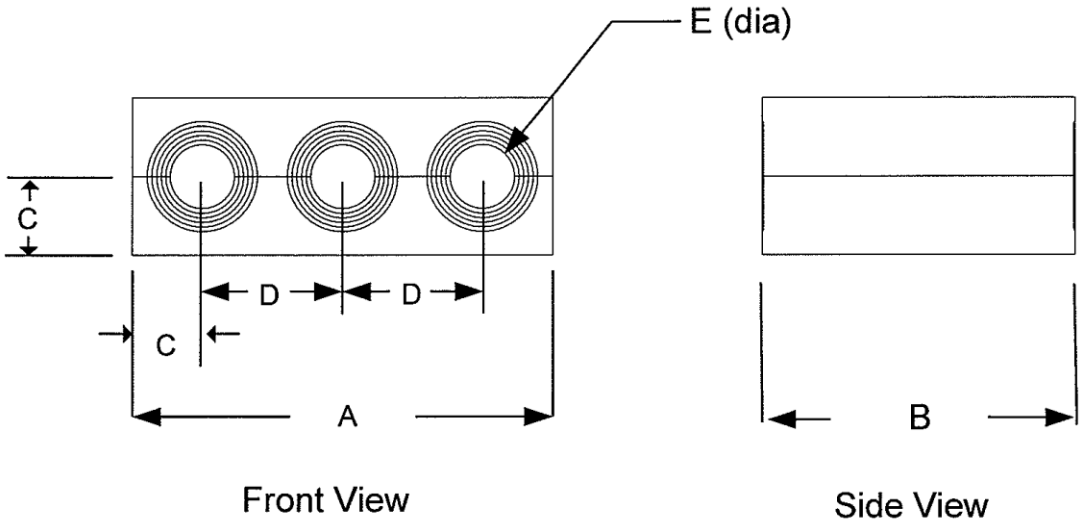
NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 4. Compact two-piece rectangular-faced two-cable insert block.

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Compact two-piece rectangular-faced three-cable insert block. Compact two-piece rectangular-faced three-cable insert blocks shall be in accordance with figure 5.



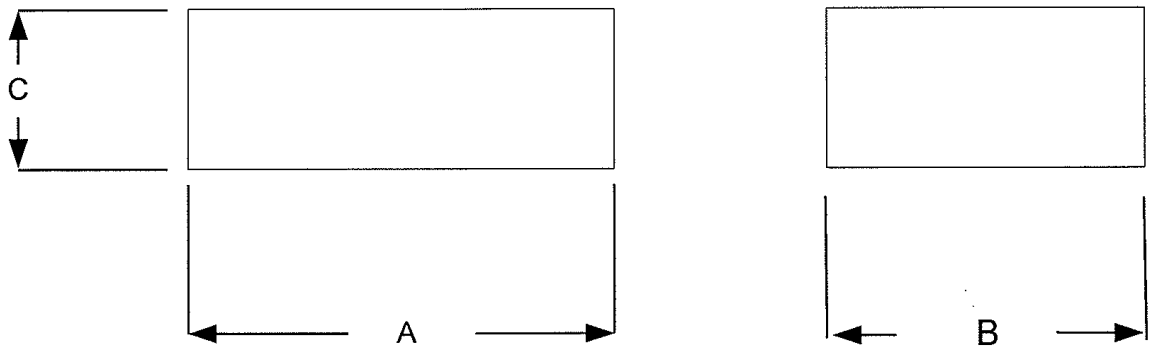
Item type	Item material	Item designation	A	B	C	D	E
B	N	15	40.0 (1.57)	30.0 (1.18)	7.5 (0.30)	12.5 (0.49)	4.0 (0.16)

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 5. Compact two-piece rectangular-faced three-cable insert block.

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Compact rectangular filler insert block. Compact rectangular filler insert blocks shall be in accordance with figure 6.



Item type	Item material	Item designation	A	B	C
I	N	10	40.0 (1.57)	30.0 (1.18)	10.0 (0.39)

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are shown in parentheses and are for information only.
3. Drawing is not to scale.
4. Item designation is an integer equal to two times the C dimension.

FIGURE 6. Compact rectangular filler insert block.

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Part or identifying number (PIN) for compact rectangular cable penetrator assembly. Compact rectangular cable penetrator components are differentiated by an item designation in accordance with table I. An example of a PIN for a compact filler insert block 5 mm high, of non-specified material, is provided below (see MIL-DTL-24705 for complete details).

M	**24705**	**/4**	**-**	**I**	**N**	**10**
Prefix for Military Specification	Specification Number	Spec Sheet No.		Item Type (see table I)	Item Material (see table I)	Item Designation (see table I)

TABLE I. PIN.

Item type	Item material	Item designation	Description	Figure
F	A	01	Compact rectangular metal frame and associated hardware	2
B	N	40	Compact two-piece square-faced one-cable insert block	3
B	N	20	Compact two-piece rectangular-faced two-cable insert block	4
B	N	15	Compact two-piece rectangular-faced three-cable insert block	5
I	N	10	Compact rectangular filler insert block	6

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REQUIREMENTS.

Requirements for compact rectangular cable penetrators are as described in table II.

TABLE II. Requirements.

	MIL-DTL-24705/4 requirement
Material, alloys, aluminum	Applicable, with the following modification: Aluminum alloy in accordance with ASTM B179 shall be used for the compact rectangular metal frame and counterframe.
Material, corrosion-resisting steel (CRES)	Applicable, with the following modification: CRES 303 shall be used for the hexagon nuts. CRES 304, in accordance with ASTM A276, shall be used for the threaded inserts.
Construction, general	The cable insert blocks shall have peelable layers to accommodate different size cables and shall be in accordance with figures 1 through 6.
Threads	Applicable, with the following modification: Metric threads in accordance with FED-STD-H28/21 and FED-STD-H28/22 shall be used. The nominal size and thread pitch shall be M6 x 1.
Vibration	Multiple cable penetrators shall show no evidence of broken, deformed, or displaced parts, chips, cracks, or physical damage when tested in accordance with 4.6.3 as modified herein.
High impact (HI) shock	Multiple cable penetrators shall show no evidence of broken, deformed, or displaced parts, chips, cracks, or physical damage when tested in accordance with 4.6.4 as modified herein.
Watertight	N/A
Nonmagnetic	When specified (see MIL-DTL-24705).
Electromagnetic compatibility (EMC)	N/A
Fire stop	N/A
Durometer	65 to 87 durometers.
Compression	5 percent (maximum) percentage change when tested in accordance with 4.6.11 as modified herein.
Expansion	1.5 percent (maximum).
Fluid immersion	The change in durometer hardness shall be not greater than ±5 percent when tested in accordance with 4.6.13 and modified herein.
Infrared (IR) signature	N/A
Spraytight	The compact rectangular cable penetrator assembly shall meet the spraytight requirement specified in MIL-STD-108 when tested as specified herein.
Fungus	Materials used in manufacturing subcomponents shall be listed as fungus inert using the guidance provided in Guideline 4 of MIL-HDBK-454, or shall be tested as specified herein. Following the test, the polymeric material shall show sparse or very restricted microbial growth and reproduction with minor or inhibited substrate utilization not exceeding MIL-STD-810 Rating I for fungus growth. There shall be little or no chemical, physical, or structural change detectable.

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TABLE II. Requirements – Continued.

	MIL-DTL-24705/4 requirement
Marking: frames and accessories	The identification marking shall be: a. Vendor Part Number (VPN). b. Manufacturer's Name or Logo. c. Military Part or Identifying Number (PIN); in lieu of being included on item, it may be included on packaging.
Marking: cable and filler insert blocks	The identification marking shall be: a. Vendor Part Number (VPN). b. Manufacturer's Name or Logo. c. Military Part or Identifying Number (PIN); in lieu of marking on item, it may be included on packaging.

VERIFICATION.

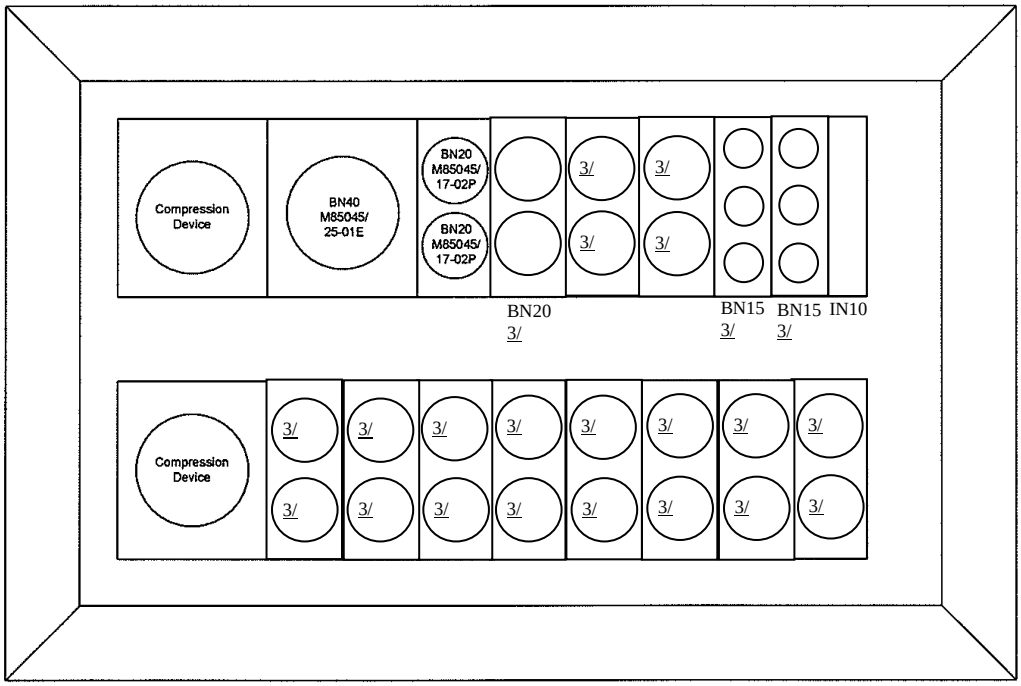
Verification methods required for compact rectangular cable penetrators are as described in table III.

TABLE III. Verification.

	MIL-DTL-24705/4 verification
Vibration	Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box. Following the test, the multiple cable penetrator test specimen shall pass the shock test in accordance with 4.6.4 as modified herein; shall pass the spraytight test specified herein; and shall show no evidence of broken, deformed, or displaced parts, chips, cracks, or physical damage; and in that order.
Shock	Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box. Following the test, the multiple cable penetrator shall pass the spraytight test specified herein.
Durometer	The sample size shall be one 40 by 40 by 60 mm solid block.
Compression	Each sample of insert material submitted for qualification shall be a 29-mm diameter by 10 mm solid block. Samples shall be tested in accordance with ASTM D395, Method B, Option 1.
Expansion	The sample size shall be one BN40 insert block.
Fluid immersion	Sample size shall be three 40 by 40 by 60 mm solid blocks. After each fluid immersion test, the samples shall be removed from the test fluid and the insert material tested shall pass test in accordance with 3.5.8.
Spraytight	The compact rectangular cable penetrator assembly shall be tested in accordance with MIL-STD-108 for spraytight. Testing may be performed with the compact rectangular cable penetrator assembly installed in a qualified MIL-DTL-24728/9 or MIL-DTL-24728/10 fiber optic interconnection box.
Fungus	Materials not listed as fungus inert in Guideline 4 of MIL-HDBK-454 shall be tested in accordance with TIA-455-56.

Qualification inspection. Qualification inspection shall consist of the examination and tests specified herein on compact rectangular cable penetrator test specimens configured as shown on figure 7. The components for the test specimen shall be as specified on figure 7. Each component listed shall be required for each test specimen.

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Item	Quantity – part/description
Frame	1-FA01
Compact two-piece cable insert blocks	1-BN40 with specified test cables
	1-BN20 with specified test cables
Compact two-piece cable insert blocks 1/	11-BN20 2/
	2-BN15 2/
Filler insert blocks	1-IN10
Test cables	1-M85045/25-01E
	2-M85045/17-02P
Lubricant	As required – C-T-91

NOTES:
1/ The compact two-piece insert blocks (BN20) can also be used as blanking insert blocks.
2/ Compact two-piece insert block with all layers and core intact.
3/ Compact two-piece cable insert block with all layers and core intact.

FIGURE 7. Qualification test specimen configurations. 3/

Qualification by similarity. Qualified manufacturers who produce cable insert blocks and filler blocks for MIL-DTL-24705/1 are qualified under this specification sheet if their product passes the size inspection, vibration, shock, and spraytight requirements specified herein, provided the only difference between the previously qualified blocks and the blocks under test is the depth of the block identified as dimension B on figures 3, 4, 5, and 6.

Conformance inspection. Expansion shall be added to MIL-DTL-24705 Group B inspection.

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APPLICABLE DOCUMENTS

General. The documents listed in this section are specified in this specification sheet.

Government documents.

Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28/21 - Screw-Thread Standards for Federal Services Section 21 Metric Screw-Threads

FED-STD-H28/22 - Screw-Thread Standards for Federal Services Section 22 Gages and Gaging for Metric Screw Threads-M and MJ Thread Forms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24705/1 - Penetrators, Multiple Cable, Electric Cable, Rectangular Metal Frame, Square-Faced Blocks

MIL-DTL-24728/9 - Interconnection Box, Cable Routing, Fiber Optic, Spraytight, 712 X 1372 mm

MIL-DTL-24728/10 - Interconnection Box, Cable Routing, Fiber Optic, Spraytight, 712 X 1017 mm

MIL-PRF-85045 - Cables, Fiber Optics, (Metric), General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-149 - Rubber

MIL-HDBK-454 - General Guidelines for Electronic Equipment

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A276 - Standard Specification for Stainless Steel Bars and Shapes

ASTM B179 - Standard Specification for Aluminum Alloys in Ingot and Molten Forms for Castings from All Casting Processes

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

(Copies of these documents are available online at www.astm.org.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-56 - Test Method for Evaluating Fungus Resistance of Optical Fiber and Cable

(Copies of this document are available online at www.tiaonline.org.)

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy – SH
DLA – CC

Preparing activity:
Navy – SH
(Project 5975-2013-003)

Review activities:
DLA – GS
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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